

Genie Space – Design Guide

Javier Poveda-Panter, Specialist Solutions Architect – AI & ML

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1. Best Practices

This chapter provides guidance on setting up effective Genie Spaces based on field experience and product recommendations.

1.1 Data Modeling

Curate Your Data

Genie works best with clean, well-structured data. Invest time in data preparation before creating a Genie Space.

Recommendations:

- **Pre-join tables:** Create denormalized views that combine frequently joined tables to reduce complexity
- **Drop unnecessary columns:** Hide or remove columns that aren't relevant to the Genie Space's purpose
- **Use descriptive naming:** Column and table names should be self-explanatory (e.g., `customer_total_revenue` instead of `cust_tot_rev`)
- **5-7 tables optimal:** Databricks recommends 5-7 tables for best accuracy (hard limit is 25, but fewer is better)
- **One domain per space:** Focus each Genie Space on a single business function (e.g., sales pipeline, HR analytics)

Data Model Approaches

Approach	Pros	Cons	Best For
Metric Views	Enforced KPI definitions, consistent calculations, semantic layer	Requires UC setup, learning curve	Strict KPI reporting
Pre-aggregated Tables	Faster queries, simpler SQL, higher accuracy	Less flexibility, requires ETL maintenance	Dashboards, standard reports
Denormalized Views	Good accuracy, simple queries, no joins needed	Data duplication, refresh complexity	General use cases

Approach	Pros	Cons	Best For
Star Schema	Familiar to analysts, flexible	Requires joins, higher hallucination risk	Ad-hoc exploration (advanced users)

When to Use Metric Views

Use **Metric Views** when consistent, governed KPI definitions are critical. Metric Views ensure that business metrics are calculated the same way every time, regardless of how the question is phrased.

1.2 Knowledge Store Configuration

The **Knowledge Store** is a collection of curated semantic definitions that enhances Genie's understanding of your data. All configurations are scoped to your Genie Space and do not affect Unity Catalog metadata.

Knowledge Store Components

Component	Purpose	Access
Space-level metadata	Table/column descriptions, synonyms	Configure > Data
Prompt matching	Format assistance, entity matching	Configure > Data > Column > Advanced
Join relationships	Define table relationships for accurate JOINS	Configure > Joins
SQL expressions	Measures, filters, dimensions	Configure > Instructions > SQL Expressions

Prompt Matching

Prompt matching helps Genie match user language to actual data values and correct spelling issues.

Format Assistance:

- Provides representative values for all eligible columns
- Helps Genie understand data types and formatting patterns
- Automatically enabled when you add tables to a space

Entity Matching:

- Provides curated lists of distinct values for categorical columns

- Supports up to **120 columns** per space
- Each column can include up to **1,024 distinct values** (max 127 chars each)
- Best for: state/country codes, product categories, status codes, department names

Example - Why Entity Matching Matters:

User asks: *"Show me car sales in Florida for Q1"*

Without Entity Matching	With Entity Matching
WHERE state ILIKE '%Florida%' (returns no results)	WHERE state = 'FL' (correct)

Tables with row filters or column masks are excluded from prompt matching. Space authors must manually disable entity matching for views that reference such tables.

SQL Expressions

SQL expressions define reusable business concepts that Genie uses when users ask about business terms.

Type	Purpose	Example
Measures	KPIs and metrics	SUM(revenue - cost) AS profit_margin
Filters	Common filtering conditions	order_value > 10000 (large orders)
Dimensions	Attributes for grouping	DATE_TRUNC('month', order_date)

Join Relationships

Define table relationships to help Genie create accurate JOIN statements:

1. Navigate to **Configure > Joins > Add**
2. Select left and right tables
3. Enter join condition (e.g., accounts.id = opportunity.accountid)
4. Select relationship type: Many to one, One to many, or One to one

When users click thumbs up on responses with joins, Genie can suggest new SQL snippets for authors to review and add to the knowledge store.

Space-Level Metadata

Customize metadata without affecting Unity Catalog:

- **Table descriptions:** Add space-specific directions for SQL generation
- **Column descriptions:** Enhance Genie's understanding of each column

- **Synonyms:** Map business terms to column names (e.g., "revenue" → `total_sales`)
- **Hide columns:** Remove irrelevant columns to keep Genie focused

References:

- [Docs: Knowledge Store Configuration](#)

1.3 Configuration Priority

When configuring a Genie Space, apply settings in this order of effectiveness:

1. **Unity Catalog Metadata** (most effective)
2. **Prompt Matching (Entity Matching + Format Assistance)**
3. **SQL Expressions**
4. **SQL Examples**
5. **Join Definitions**
6. **General Instructions** (least effective)

1.4 Writing Effective Instructions

Key Insight: One high-quality SQL example teaches Genie more effectively than 10 lines of text instructions. Focused, domain-specific spaces outperform broad implementations by 60-80% in accuracy.

Core Principles

1. **Domain Specificity Over Breadth** - One business function per space, 5-7 tables maximum
2. **Example-Driven Learning** - Concrete SQL examples > Abstract text explanations
3. **Token Optimization** - Keep instructions under 2,000 characters

Token Optimization Techniques

Technique	Impact
Bullet points vs sentences	40% reduction
Abbreviations after first use	25% reduction
Table-based mappings	35% more efficient

References:

- Document: Writing Effective Genie Instructions

1.5 Testing & Validation

Use Genie's benchmarking feature to measure and track accuracy. [GenieIQ](#) is a community tool that scores Genie Spaces (0-100) and provides specific improvement recommendations.

1.6 API Integration Best Practices

- **Implement exponential backoff** for rate limit handling (429 errors)
 - **Use polling** to check for query completion rather than blocking
 - **Cache responses** where appropriate to reduce API calls
 - **Handle timeouts gracefully**—Genie queries can take 30+ seconds
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2. Limitations

This chapter outlines the known limitations of the Databricks Genie Agent.

2.1 Preview Limitations

Query Rate Limits

Each **workspace** is limited to **20 questions per minute** (UI) or **5 QPM** (API free tier) **across all Genie Spaces**. This is a workspace-level limit, not per-Space.

References:

- [Docs: Throughput Limit](#)

Capacity Limits

Each Genie Space can support up to **10,000 conversations**, and each conversation can include up to **10,000 messages**.

Why this matters: For high-volume deployments, these limits may eventually be reached. However, for most use cases, these limits are generous.

References:

- [Docs: Technical Requirements and Limits](#)

2.2 Design Limitations

Table Limit

A single Genie Space supports a maximum of **25 tables or views**. However, **5-7 tables is recommended** for optimal accuracy.

Why this matters: Genie needs to understand table relationships, column semantics, and data types to generate accurate SQL. With more than 25 tables, the context window becomes too large, leading to increased hallucinations and incorrect table/column selection.

Mitigation:

- Split tables across multiple Genie Spaces by data domain (e.g., stock data, manufacturing, sales)
- Optimize each Genie Space for its specific domain with targeted instructions
- Combine Genie Spaces using the **AgentBricks Agent Orchestrator** for cross-domain queries
- Consider creating denormalized views that combine frequently joined tables

References:

- [Docs: Technical Requirements and Limits](#)

SQL Example Limit

Genie Spaces are limited to **100 SQL examples** in instructions. (SQL snippets and JOIN relationships limit increased to 200)

Why this matters: SQL examples are the primary way to teach Genie domain-specific query patterns, business logic, and preferred SQL syntax. Customers with complex domains often want more examples.

How Genie uses examples: Genie intelligently selects relevant SQL examples—it doesn't load all 100 into context. This means well-crafted, diverse examples are more valuable than quantity.

Mitigation:

- Focus on high-value, representative examples that cover common query patterns
- Use clear, descriptive column/table naming to reduce the need for examples
- Add business logic explanations in table/column comments in Unity Catalog
- Prioritize examples for complex calculations and domain-specific terminology

References:

- [Docs: How Genie generates a response](#)

Entity Matching Limits

Entity matching supports up to **120 columns** per space, with each column limited to **1,024 distinct values** (max 127 characters each).

Exclusions:

- Tables with **row filters** or **column masks** are automatically excluded from prompt matching
- Space authors must manually disable entity matching for **views that reference filtered tables** or **dynamic views**

References:

- [Docs: Knowledge Store - Entity Matching](#)

Compute Requirements

Genie spaces require a **Pro or Serverless SQL warehouse**. You must have at least **CAN USE** permission on the selected warehouse. If this permission is missing, space creation fails.

References:

- [Docs: Technical Requirements and Limits](#)
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Documentation

- [Genie Best Practices \(Public Docs\)](#) - Official best practices
- [Genie Setup Guide](#) - Technical requirements and setup
- [Knowledge Store Configuration](#) - Value dictionaries and data sampling